PART 1 - GENERAL

1.1 Summary

26 05 00 Part 1.1.1.A This section specifies the general electrical requirements common to all electrical specification sections for the Meridian Data Centre Campus, Phase 1, located in Navi Mumbai, India. All electrical work shall comply with the requirements of this section and the applicable individual equipment sections.

26 05 00 Part 1.1.1.B Related specification sections include but are not limited to: Section 26 33 53 Static Uninterruptible Power Supply, Section 26 32 13 Engine Generators, Section 26 13 26 Medium Voltage Switchgear, Section 23 81 23 Computer Room Air Handling Units, and Section 21 22 00 Clean Agent Fire Suppression Systems.

1.2 References

26 05 00 Part 1.2.1.A The following Indian and International standards form part of this specification to the extent referenced herein: IS 732 (Code of Practice for Electrical Wiring Installations), IS 3043 (Code of Practice for Earthing), IS 694 (PVC Insulated Cables), IEC 60364 (Low Voltage Electrical Installations), IEC 61439 (Low-Voltage Switchgear and Controlgear Assemblies).

26 05 00 Part 1.2.1.B All electrical installations shall comply with the National Electrical Code (NEC) of India, the National Building Code (NBC) of India Part 8 Building Services Section 2 Electrical and Allied Installations, and the local regulations of the Maharashtra State Electricity Distribution Company (MSEDCL).

1.3 Submittal Requirements

26 05 00 Part 1.3.1.A The Contractor shall submit equipment shop drawings, product data sheets, and compliance statements for each electrical equipment item specified in the individual equipment sections. Submittals shall be transmitted using the project standard transmittal form with a unique submittal identification number.

26 05 00 Part 1.3.1.B Each equipment submittal shall include manufacturer's published product data, certified performance test reports, dimensional drawings, wiring diagrams, and a clause-by-clause compliance matrix referencing the applicable spec section.

26 05 00 Part 1.3.1.C Submittal review turnaround time is 14 calendar days from receipt of the complete package. Incomplete submittals will be returned without review.

1.4 Quality Assurance

26 05 00 Part 1.4.1.A Equipment manufacturers shall have a minimum of 10 years demonstrated experience in the design and manufacture of the specified equipment type, with verifiable installations in mission-critical data centre facilities.

26 05 00 Part 1.4.1.B Equipment manufacturers shall hold current ISO 9001 quality management system certification from an accredited certification body for all manufacturing facilities producing equipment for this project.

26 05 00 Part 1.4.2.A The installing contractor shall be licensed by the appropriate state regulatory authority and shall have a minimum of 5 years experience in installing similar equipment in mission-critical or Tier III and above data centre environments.

26 05 00 Part 1.4.3.A All factory-assembled electrical equipment shall be subjected to routine tests at the manufacturer's facility prior to shipment. Factory test certificates shall be submitted as part of the equipment submittal package.

26 05 00 Part 1.4.3.B All medium voltage equipment and switchgear assemblies shall be type tested and certified in accordance with the applicable IEC standards. IEC type test certificates from an accredited third-party laboratory (ASTA, KEMA, or equivalent NABL-accredited body) shall be provided for each equipment type.

26 05 00 Part 1.4.3.C Low voltage switchgear and controlgear assemblies shall be type tested in accordance with IEC 61439-1 and IEC 61439-2. Design verification reports shall be submitted for Consultant review.

26 05 00 Part 1.4.3.D Power cables rated above 1.1 kV shall be type tested to IS 7098 Part 2 or IEC 60502-2 as applicable. Cable test certificates shall include conductor resistance, insulation resistance, and high voltage withstand test results.

26 05 00 Part 1.4.3.E Current transformers and voltage transformers shall be type tested in accordance with IEC 61869-2 and IEC 61869-3 respectively. Accuracy class certificates shall be submitted.

26 05 00 Part 1.4.3.F Protective relays and metering equipment shall be factory-calibrated and supplied with calibration certificates traceable to national standards. Relay settings shall be submitted for Consultant approval prior to dispatch.

26 05 00 Part 1.4.3.G Earthing and lightning protection system components shall comply with IS 3043 and IEC 62305. Material test certificates for copper earthing conductors and earth electrodes shall be submitted.

26 05 00 Part 1.4.3.H Emergency lighting and exit signs shall be tested to IS 10118 and IS 6665. Battery backup duration and illumination level test reports shall be submitted.

26 05 00 Part 1.4.3.I Fire detection and alarm system components shall be tested and listed to IS 2189 or UL 268 / UL 864 as applicable. Third-party listing certificates shall be submitted.

26 05 00 Part 1.4.3.J Diesel generator auxiliary systems including fuel transfer pumps, day tank level controls, and exhaust treatment systems shall be factory tested per the applicable IS or IEC standards. Test records shall be submitted.

1.5 Warranty

26 05 00 Part 1.5.1.A The Contractor shall provide a minimum warranty period of 24 months from the date of substantial completion or 30 months from the date of delivery, whichever occurs first, for all electrical equipment and installations.

PART 2 - PRODUCTS

2.1 General Electrical Requirements

26 05 00 Part 2.1.1.A The campus electrical distribution system shall be designed for a nominal medium voltage of 11 kV, 3-phase, 50 Hz from the utility incoming feed and standby generator plant. Secondary distribution shall be at 415 V, 3-phase, 4-wire plus ground, 50 Hz.

26 05 00 Part 2.1.1.B Power factor at the point of common coupling shall be maintained at a minimum of 0.95 lagging. Automatic power factor correction equipment shall be provided where required.

PART 3 - EXECUTION

3.1 Examination and Preparation

26 05 00 Part 3.1.1.A Prior to installation, the Contractor shall examine all areas and conditions under which electrical equipment is to be installed and notify the Consultant in writing of any conditions that may adversely affect the installation.

3.2 Installation

26 05 00 Part 3.2.1.A All electrical equipment shall be installed in strict accordance with the manufacturer's written installation instructions, approved shop drawings, and the requirements of this specification.